

Data & AI Unit

Document Metadata

Field	Detail
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Software Design Document (SDD) Template

1. Introduction

- **Purpose:** [Describe the purpose of this document. E.g., to define the design of the XYZ system.]
- **Scope:** [Summarize the system's objectives and what is in/out of scope.]
- **Definitions and Acronyms:** [List and define important terms.]
- **References:** [Link to related documents: requirements, API specs, etc.]

2. System Overview

- **System Description:** [High-level overview of the system.]
- **Design Goals:** [E.g., scalability, maintainability, security.]
- **Architecture Summary:** [Monolith, microservices, serverless, etc.]

3. Architectural Design

- **System Architecture Diagram:**
 - *Use Mermaid diagram here.*
- **Module Breakdown:**
 - [Module 1]: [Responsibilities, interactions.]
 - [Module 2]: [Responsibilities, interactions.]
- **Technology Stack:** [Languages, frameworks, databases.]
- **Data Flow and Control Flow:**
 - *Use Mermaid sequence or flow diagrams here.*

4. Detailed Design

For each module:

[Module Name]

- **Responsibilities:** [What does it do?]
- **Interfaces/APIs:**

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- Inputs: [Describe input data.]
- Outputs: [Describe output data.]
- Error Handling: [Describe approach.]
- **Data Structures:** [Key models/schemas.]
- **Algorithms/Logic:** [Design patterns or important logic.]
- **State Management:** [How is state handled?]

5. Database Design

- **Entity-Relationship (ER) Diagram / Schema Diagram:**
 - *Use Mermaid ER diagram here.*
- **Tables/Collections:** [Define each with fields and constraints.]
- **Relationships:** [Describe relationships between entities.]
- **Migration Strategy:** [If applicable.]

6. External Interfaces

- **User Interface:** [Mockups, UX notes.]
- **External APIs:** [Integrations and dependencies.]
- **Network Protocols/Communication:**
 - REST (Representational State Transfer), GraphQL, gRPC (Google Remote Procedure Call) , etc.

7. Security Considerations

- **Authentication:** [Method used.]
- **Authorization:** [Role/permission models.]
- **Data Protection:** [Encryption, storage.]
- **Compliance:** [GDPR, HIPAA, Data Protection Act, EU AI Act etc.]
- **Threat Model:**
 - *Use Mermaid diagram here if helpful.*

8. Performance and Scalability

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- **Expected Load:** [Requests per second, data volume.]
- **Caching Strategy:** [Describe caches used.]
- **Database Optimization:** [Indexes, partitioning.]

9. Deployment Architecture

- **Environments:** [Dev, staging, production.]
- **CI/CD Pipeline:** [Tools and stages.]
- **Cloud/Hosting:** [AWS, GCP, Azure, etc.]
- **Containerization/Orchestration:** [Docker, Kubernetes.]

10. Testing Strategy

- **Unit Testing:** [Tools, coverage goals.]
- **Integration Testing:** [Approach and tools.]
- **End-to-End Testing:** [Scope and tools.]
- **Quality Metrics:** [Code coverage, linting, etc.]

11. Appendices

- **Diagrams:** [All referenced diagrams.]
- **Change History:**
 - [Version, Date, Author, Changes]